

Python List Operations

1. Proper Program Prompt

Write a Python program to demonstrate common list operations using cricket player names and jersey numbers.

The program should:

1. Create a list of cricket players.
2. Add players using `append()`.
3. Insert a player at a specific position using `insert()`.
4. Remove a player using `remove()`.
5. Remove an element at a specific index using `pop()`.
6. Find the index of a player using `index()`.
7. Count the occurrences of a player using `count()`.
8. Create a list of jersey numbers and sort it using `sort()`.
9. Reverse the sorted list using `reverse()`.
10. Copy a list using `copy()` and display both the original and copied lists.
11. Find the maximum value, minimum value, and number of elements using `max()`, `min()`, and `len()`.
12. Remove all elements from a list using `clear()`.
13. Print the list after each operation so that the result of each operation can be observed.

2. Python Program

```
players = ["virat", "dhoni", "rohit", "jadeja"]

players.append("bumrah")
print(players)

players.append("kl rahul")
print(players)

players.insert(2, "dravid")
print(players)

players.remove("rohit")
print(players)

players.pop(3)
print(players)

print(players.index("virat"))
players.append("dhoni")
print(players.count("dhoni"))

players2 = [18, 7, 19, 97, 1]

players2.sort()
print("Sorted list:", players2)

players2.reverse()
print("Reversed list:", players2)

players = ["virat", "dhoni", "rohit", "jadeja"]

players2 = players.copy()

print("Original list:", players)
print("Copied list:", players2)
```

```
players2 = [18, 7, 19, 97, 1]

print("List:", players2)

print("Maximum value:", max(players2))
print("Minimum value:", min(players2))
print("Number of elements:", len(players2))

players2.clear()

print("After clear():", players2)
```

3. Sample Output

```
['virat', 'dhoni', 'rohit', 'jadeja', 'bumrah']
['virat', 'dhoni', 'rohit', 'jadeja', 'bumrah', 'kl rahul']
['virat', 'dhoni', 'dravid', 'rohit', 'jadeja', 'bumrah', 'kl rahul']
['virat', 'dhoni', 'dravid', 'jadeja', 'bumrah', 'kl rahul']
['virat', 'dhoni', 'dravid', 'bumrah', 'kl rahul']
0
2
Sorted list: [1, 7, 18, 19, 97]
Reversed list: [97, 19, 18, 7, 1]
Original list: ['virat', 'dhoni', 'rohit', 'jadeja']
Copied list: ['virat', 'dhoni', 'rohit', 'jadeja']
List: [18, 7, 19, 97, 1]
Maximum value: 97
Minimum value: 1
Number of elements: 5
After clear(): []
```